A Source-Agnostic AI-Powered Smart City Observatory: Self-Generating, Self-Healing, and Self-Deploying Data Wrappers in Production

Anonymous Author(s)

***Abstract*—An urban observatory is worth as many indicators as it can actually take in, and that number is usually small. Every feed, a statistics-office API here, a town spreadsheet there, an occasional CSV, arrives in its own shape and needs its own extraction code that somebody writes and then keeps alive. We describe the Observatory, a smart city observatory in production whose intake layer removes that per-feed labour: a non-technical user registers a source through a form, and the platform then writes, runs, and maintains the extraction code on its own, whatever the source looks like. The design idea is to treat each source's wrapper as a self-managing component driven by a single autonomic control loop, in the monitor–analyze–plan–execute–knowledge (MAPE-K) sense, whose actuator is a large language model that emits and revises the wrapper's code. Three behaviours follow from that one loop. The model writes a wrapper by completing a fixed template from the source's metadata and a small sample, querying the live endpoint first when the source is an API. The wrapper is then launched, supervised, and check-pointed as an isolated process with nobody watching. And when it later throws, the loop returns the crash trace to the model and ships a corrected version. Over 149 days the platform stood up 396 wrappers across API, spreadsheet, and CSV feeds and gathered 164,322 measurements. Of the 43 wrappers (10.9 %) that crashed and were auto-repaired, 39 ended in a healthy state and 32 went back to delivering data, each repair taking a median 33.4 s; the entire stack fits in roughly 1.5 GiB. The pattern has a clear edge, too: defects in the generated code and a source's formatting quirks lie within the loop's reach, whereas spent quotas, wrong credentials, and badly set limits are environmental and lie beyond it.**

***Index Terms*—smart city observatory, source-agnostic data ingestion, autonomic computing, self-healing systems, LLM code generation, ETL automation, smart tourism**

I. Introduction

Running a city on evidence means tracking many numbers at once, and those numbers neither sit in one place nor arrive in one form. Air and water quality come off environmental sensors and monitoring APIs; overnight stays and lodging capacity come from tourism boards; energy use, mobility counts, and cultural attendance come from utilities, transport operators, and municipal departments. The publishers differ and so do the ways in to their data, a REST endpoint here, an Excel workbook or a CSV export there, along with the field names, the way dates are written, and the language they are written in. None of it stands still, either: an indicator earns its place only while it stays current, so every feed has to be pulled again and again as new readings land, and every upstream change of format or endpoint has to be taken in without breaking the series a city has already built up. What an observatory confronts is therefore a standing integration problem over many heterogeneous sources, not a task that is finished once.

The usual way to take a source in is to write an extraction wrapper for it and then keep that wrapper alive as the source shifts underneath. This cost, paid once per source to build and then indefinitely to maintain, is why mediator-style integration has never reached past a small set of feeds [1]–[3], and it sets the ceiling on coverage: an observatory ends up showing whichever feeds someone found time to wire and to nurse, rather than the city.

We call an intake layer *source-agnostic* when onboarding a feed costs no bespoke code: a domain user completes a registration form, and the platform carries the source from there through extraction and continued upkeep without further human work. This paper reports the Observatory (anonymized for review), a smart city observatory running for a European region, where source-agnostic intake is obtained by making each wrapper an *autonomic* element in the sense of Kephart and Chess [4]: a component that manages itself through a closed loop over its own lifecycle. What is unusual is the actuator. A conventional autonomic manager adjusts the knobs of a fixed component; ours asks a large language model (LLM) to write the component's code, and to rewrite it after a failure.

We organize the wrapper's three self-* behaviours as the monitor–analyze–plan–execute loop over shared knowledge (MAPE-K) of the autonomic-computing blueprint [5], [6]. At registration, *self-generation* (Section IV-B) plans the artifact: the LLM completes a fixed template from the indicator's metadata and a data sample, interrogating a live API through a controlled fetch tool before it writes anything. *Self-deployment* (Section IV-C) is the execute step that runs unattended: each wrapper is an isolated process the platform starts, watches, check-points, and resumes from per-wrapper watermarks after any restart. *Self-healing* (Section IV-D) is the loop closing on a live artifact: a crash is detected, the termination is classified, the model re-plans from the captured trace, and the corrected wrapper goes back out.

This paper frames the system as a source-agnostic auto-

nomic manager and measures it in production. Three contributions follow. We set out the observatory's architecture and its one narrow contract: every source is reduced to a stream of normalized $(\text{timestamp}, \text{value})$ points by a wrapper the platform owns end to end. That contract is what makes intake source-agnostic (Section III). We then unify the three self-* behaviours as a single autonomic manager over a shared knowledge base, assign each its MAPE-K role, and position it against autonomic computing, LLM-based code generation and repair, and wrapper maintenance (Section IV; related work in Section II). Finally we report from the running deployment over 149 days, using a pre-automation reference period and a fault analysis that fixes the manager's scope of control (Section V).[1]

II. Background and Related Work

A. Smart city and tourism observatories

The smart-tourism literature has long argued for destination-scale data infrastructure [7], [8] and has codified the indicators worth tracking through schemes such as ETIS [9], while field reports describe patchy uptake [10] and dashboards assembled mostly from warehouses and surveys [11]. A recurring finding in evaluations of smart-destination initiatives is that the binding constraint on sustained monitoring is getting the data in at all [12], [13]. These accounts say little about how intake is actually built. Our claim is that the intake layer, specifically its ability to absorb arbitrary sources, sets the ceiling on how many indicators an observatory can keep alive, and we report one such layer from production. The deployment happens to be a tourism observatory, but nothing in the design depends on that: it needs only heterogeneous sources and a shared time-series target.

B. Autonomic computing and self-* systems

Autonomic computing proposed self-managing components with closed control loops as the response to operational complexity [4], structured as a monitor–analyze–plan–execute cycle around a shared knowledge base (MAPE-K) [5], and later surveys catalogue its self-* property families [6]. the Observatory applies that blueprint to data wrappers, but swaps the actuator: instead of tuning a fixed component's parameters, the manager regenerates the component's source code through an LLM, so configuration becomes synthesis and recovery becomes repair.

[1]For review, the platform's services are mirrored anonymously on Anonymous GitHub: the orchestrator (https://anonymous.4open.science/r/tsd-orchestrator-84B0), the resource service (https://anonymous.4open.science/r/tsd-resource-service-D9C4), the indicator service (https://anonymous.4open.science/r/tsd-indicator-service-7A1F), the data collector (https://anonymous.4open.science/r/tsd-data-collector-3B82), the user service (https://anonymous.4open.science/r/tsd-user-service-C641), the user interface (https://anonymous.4open.science/r/tsd-user-interface-5E9A), and the knowledge base (https://anonymous.4open.science/r/tsd-knowledge-base-2F47); the scripts that compute the reported numbers accompany the submission.

C. LLMs for code generation, repair, and extraction

Language models now write runnable programs from prose [14], and automated repair has moved toward LLMs as well [15]. Several systems close a feedback loop around the generated code, whether through self-critique [16], episodic reflection [17], or conditioning on execution results [18], though the benefit of self-repair is inconsistent [19]. On the data side, Evaporate established that synthesizing extraction code is cheaper at scale than calling a model per record [20], AutoScraper builds reusable scrapers with an agent [21], and both prompted [22] and fine-tuned [23] models handle textbook preprocessing steps. Our planner reuses these ideas; what differs is the signal and the unit. The signal that prompts a plan is a crash from production rather than a failing unit test, and the plan rewrites the entire managed artifact instead of editing a faulting line. We do not claim novelty in the code generation itself; the contribution is the autonomic-systems framing around it and the longitudinal production evidence, neither of which the systems above set out to provide.

D. Wrapper induction, maintenance, and ETL automation

Early wrapper induction derived extraction rules from labelled examples [2], later systems learned them without supervision from structural regularities [24], and the tooling has been surveyed at length [25], [26]. Keeping a wrapper working as its source drifts, by spotting the break and re-deriving the rules, was identified as a distinct problem from the start [3], [27]. ETL pipelines, meanwhile, have mostly been designed by hand rather than generated [28], [29], with automation reaching adjacent pieces such as schema matching [30], large-scale curation [31], and wrangling [32]. the Observatory folds both the initial derivation and the later re-derivation into LLM code generation, which reaches past template-structured pages to arbitrary APIs and workbooks, and it targets exactly the slice an indicator platform needs: pulling one series from one source into one normalized schema.

III. A Source-Agnostic Observatory

The Observatory runs as a set of containerized services, user, knowledge-base, data-collector, resource, and indicator (all Python and FastAPI), reached through an nginx gateway and fronted by a React dashboard. Control traffic and data traffic ride separate RabbitMQ brokers, and the resource and indicator services each keep their own datastore, with an optional Keycloak provider for single sign-on. Figure 1 lays out these components and the two planes that join them: a REST serving plane through the gateway, and an asynchronous ingestion plane across the brokers. Figure 2 then traces the data path through them.

Source-agnostic intake rests on one deliberately thin interface. Regardless of where its data comes from, a wrapper emits point batches $\{(x, y)\}$ in which x is an ISO 8601 timestamp and y a floating-point value; the code above that interface is generated per source, and everything below it is shared, fixed plumbing. Registration is a web wizard: the user either uploads a spreadsheet or CSV file or points the platform at an

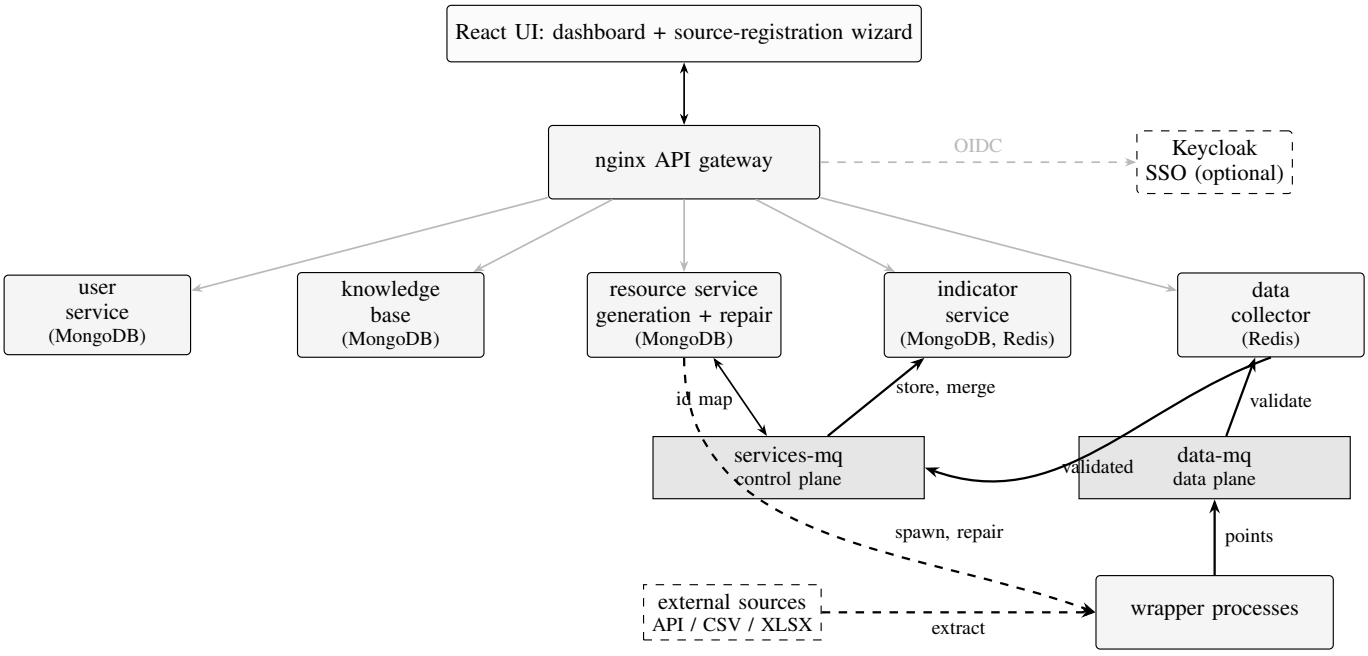


Fig. 1. Whole-system architecture. The React front end reaches the Python/FastAPI services through the nginx gateway (the REST serving plane); ingestion runs asynchronously across two RabbitMQ brokers, a data plane and a control plane, with wrapper processes that the resource service spawns and repairs pulling from the external sources. Each service keeps its own datastore, and an optional Keycloak provider handles single sign-on.

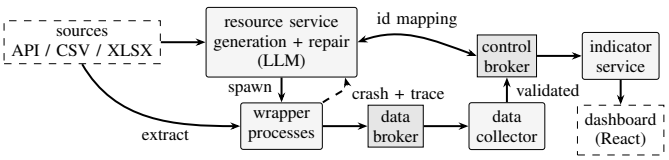


Fig. 2. Data path through the Observatory. Wrappers are generated, launched, and repaired by the resource service; the points they emit cross two brokers on their way to the indicator store and the dashboard.

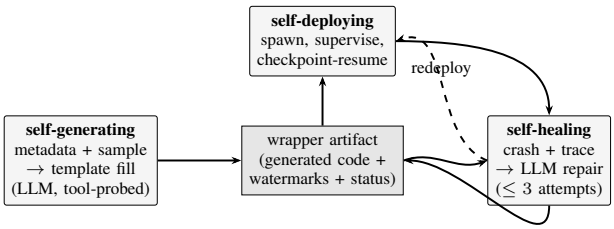


Fig. 3. One manager, three loops. Each acts on the same wrapper artifact; the actuator is an LLM that writes and revises code together with a supervisor that runs the result.

API endpoint and its credentials, and supplies the indicator's metadata, name, domain, unit, periodicity, and a description. The resource service then takes ownership of that wrapper for life. On the data side, the collector checks each batch against the interface, keeps the most recent batch per wrapper, and passes valid points to the control broker; there the resource service translates wrapper identifiers into durable resource identifiers, and the indicator service persists the segments, stitches them into ordered series, and answers the dashboard's queries. Because source type is one argument to a single generation routine rather than a code path of its own, the same machinery extends to a new kind of feed without new code.

IV. Self-* Wrappers as an Autonomic Manager

Each wrapper is the element a single autonomic manager looks after. That element is the generated artifact plus the state that travels with it: the wrapper's code, its execution watermarks, its retry counter, and its status. The platform turns the MAPE-K loop [5] on that element, watching process state, classifying how a process ended, planning a fresh or corrected artifact, and carrying the plan out by deploying it, all against a shared store of knowledge (Table I). The three self-* loops in Figure 3 are what this decomposition yields. In every one of them the actuator is the same pairing: an LLM that produces and revises code, and a process supervisor that puts it into service.

A. Shared knowledge

What binds the loops into one manager, rather than three disconnected mechanisms, is the knowledge they share. Some of it is fixed. The wrapper template encodes the Section III contract as code, and a standing list of fourteen hard rules captures lessons from earlier failures, among them: leave the historical-backfill driver alone, invent no data, treat date ranges as inclusive, keep timezone handling uniform, and read non-English dates such as localized month names. Together these constrain what the planner is allowed to emit. The rest is per-wrapper and changes at runtime: a pair of watermarks (the earliest and latest timestamps sent) with a phase, a retry

TABLE I
THE AUTONOMIC LOOP MAPPED ONTO THE PLATFORM. ONE MANAGER REALIZES ALL THREE SELF-* PROPERTIES; THE ACTUATOR IN THE *plan* STEP IS AN LLM.

MAPE-K stage	Realization in the Observatory
Monitor	30 s poll of each wrapper subprocess; exit codes, watchdog terminations, last-data timestamps
Analyze	classify a termination: code crash (exit 1) vs. idle watchdog timeout vs. provider/quota error; only crashes route to repair
Plan	LLM generates or repairs the whole wrapper file, tool-probing live APIs, under the fixed template and hard rules
Execute	static validation, then spawn or relaunch the isolated process, resuming from watermarks; semaphore-bounded
Knowledge	fixed template and the rule list distilled from past failures; per-wrapper watermarks, retry counter, execution log, status

counter capped at three, a timestamped execution log, and the stored wrapper document carrying its metadata, source configuration, current code, and status. The generation loop writes the first version of the code, the deployment loop reads and advances the watermarks, and the healing loop reads the code and the log and writes a successor. The rule list is the part that grows: each diagnosed failure can add a constraint that every later plan must respect.

B. Self-generation: planning a new artifact

The actuator is Google's Gemini Flash, reached through the `google-genai` SDK and pinned to the floating `gemini-flash-latest` alias. It does not write code freely; it works inside a fixed wrapper class (Figure 4). Only a few methods are open to it, extraction (`fetch_external_data`) and, for APIs, a windowed fetch and a polling period, while the backfill driver, the message transport, the batching, and the run loop are library code the prompt tells it not to touch (Figure 5). Jinja2 assembles the prompt. Into it go the indicator metadata, the source configuration, and a brief sample of the data: the opening 20 lines of a CSV, 15 rows from each workbook sheet alongside the sheet names, or a captured API reply. Alongside the sample sit the required `{x,y}` output shape, the template, instructions specific to the source type, and the rule list. When the source is an API, the same call hands the model a sandboxed `fetch_url` tool that returns up to 2500 characters of a live response; it may use the tool as many as fifteen times to learn the response's nesting, its field names, and which date filters the endpoint honours, rather than inferring them from the URL. Request authentication is built into the headers by the service from the declared scheme, so the model never handles it. A static check gates deployment, parsing the file and looking for placeholders left unfilled; a first failure earns one corrective prompt, a second sends the wrapper to `error`. Nothing is executed as a test beforehand, so any runtime fault is the healing loop's problem.

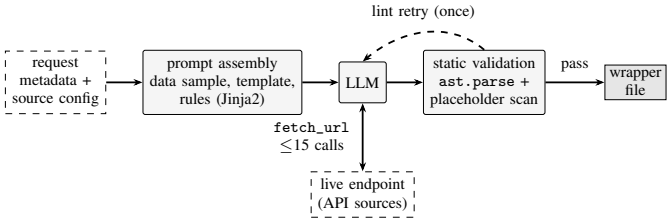


Fig. 4. The generation path. Metadata, a sample, the template, and the rules go into the prompt; for an API the model first probes the live endpoint; and the result must pass the static check, with a single retry, before it is launched.

```
class Wrapper:
    # fixed: transport, backfill, run loop (do not edit)
    async def fetch_historical_data(self):
        return await self.backfill(
            self.fetch_date_range, self.send_to_queue)

    # variable: the model fills only these
    async def fetch_external_data(self):
        """return [{'x': <ISO-8601>, 'y': <float>}, ...]"""
        ... # source-specific extraction
    async def fetch_date_range(self, start, end):
        ... # API only: windowed fetch
    def get_interval_seconds(self):
        ... # API only: poll period
```

Fig. 5. The wrapper contract (shortened). Only the open methods are the model's to fill; backfill, transport, and scheduling are fixed. Holding generation to the extraction logic is what reduces source type to an argument instead of a codebase.

C. Self-deployment: execute without an operator

Nothing about putting a wrapper into production needs a person. The resource service runs each one as a separate operating-system process, passing the wrapper identifier and source configuration on the command line and the broker URL in the environment. The number of processes alive at once is held under a semaphore (defaulting to 100), and a watchdog (defaulting to 3600 s) caps any single run. A file wrapper executes once and exits; an API wrapper first back-fills history in size-adapting batches and then settles into polling at the period its indicator's periodicity implies (hourly turns into 3600 s, daily into 86 400 s), the loop living inside the process with no external scheduler. Output leaves in batches of up to 1000 points. A monitor reads each process's state on a 30 s cycle. Every wrapper keeps a low and a high watermark, its earliest and latest emitted timestamps; after a restart it picks up from the high mark instead of re-fetching, which keeps a relaunch from either duplicating or skipping points, and the resume position is handed back through the environment. Above the individual wrappers, a container updater pulls fresh service images into production with no manual rollout. This execute step is what lets the other two loops run on their own: generation produces a process to oversee, and healing produces a replacement to relaunch and resume.

D. Self-healing: the full loop on a live artifact

Healing is where every MAPE-K stage shows up at once. The 30 s monitor reads each process's exit status (*monitor*). An abnormal exit, in practice always code 1, is separated

from an idle watchdog kill and from a provider error during generation, and only the first becomes a repair (*analyze*). The plan is an LLM call carrying the full crashed source, its captured stderr, and the final twenty lines of stdout, with the instruction to return a complete corrected file that preserves the class and addresses that particular traceback (*plan*); for an API wrapper it may re-interrogate the endpoint through the same `fetch_url` tool, up to ten times. The returned file goes through the same static check as a first generation, replaces the old file, bumps the retry counter, and the process is relaunched and resumed from its watermarks (*execute*). The manager grants at most three repairs to any wrapper and leaves the fourth-time loser in `error` for a person. Two things come from treating the artifact, not a diff, as the unit of repair. The trigger is a live exit code that may arrive weeks after generation, so the feedback carries the source's real behaviour rather than a test author's guess; and because the plan re-emits the whole wrapper, a repair can re-shape the extraction end to end instead of touching one statement [18], [19].

E. Scope of control

Source-agnosticism then comes out of the manager rather than from any separate component. One knowledge base and one set of loops cover API, spreadsheet, and CSV feeds because source type is a parameter the template takes, so a new kind of feed is absorbed simply by generating, deploying, and repairing its wrapper with no human writing code. The same view marks the limit of the manager's reach. Faults that live in the managed element, broken extraction logic and the idiosyncrasies of a source's format, are inside its authority and are dissolved by generation and repair; faults in the surroundings, a spent model quota, a stale credential, a watchdog set shorter than a source's polling period, are outside it. We return to that line in Section V with the production fault counts, and note that it is partly a design choice: some "environmental" faults, the poll-interval mismatch in particular, are values the generator itself sets and could in principle be brought inside the loop.

V. Production Deployment and Metrics

The figures here come from a 149-day run (January 13, 2026 to June 11, 2026) captured as one complete state snapshot: the wrapper collection (396 documents, generated code and execution logs included), the per-wrapper logs, the resource and indicator stores, the service logs, and container statistics; a script bundled with the submission computes every number below. At the snapshot the deployment spanned 17 containers and about 2,082 MiB of resident memory while idle, dropping to roughly 1.5 GiB once the optional single-sign-on container is set aside; the resource service, host to every wrapper process, was the heaviest at 682 MiB.

A. Generation and coverage

The manager produced 396 wrappers, 188 for spreadsheet sources, 184 for APIs, and 24 for CSV, distributed over the three domains (environment 203, economy 139, sociocultural

TABLE II
Per-source-type outcomes over the 149-day run. *Recovered* counts wrappers that went through the repair loop and closed in a non-error state. One set of loops handles every source type.

Type	n	Completed	Crashed	Recovered	Points
XLSX	188	104 (55.3%)	26	26	53,530
API	184	52 (28.3%)[a]	14	10	79,411
CSV	24	14 (58.3%)	3	3	31,381
All	396	170 (42.9%)	43	39	164,322

[a]Continuous API pollers seldom finish on their own; 54 reached `error` when the watchdog stopped them mid-idle (Section V-D), which is not a code fault.

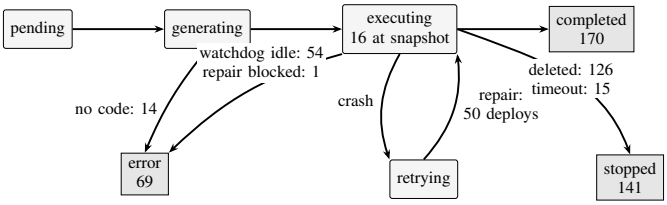


Fig. 6. Status machine, annotated with production counts ($n = 396$). The crash-and-repair branch touched 43 wrappers across 50 redeployments; most `error` outcomes are watchdog kills of idle API pollers rather than faulty code (Section V-D).

54); between them they collected 164,322 points (53,530 from spreadsheets, 79,411 from APIs, 31,381 from CSV). Volume is long-tailed: the median wrapper contributed 27.5 points and the 90th percentile 967, with 284 of 396 sending at least one. The time from a generation request to the first running process had a median of 26.8 s and a 90th percentile of 109.8 s over 331 wrappers, a figure that folds in queueing, sampling, the model call, and the static check. Further down, 118 of the 179 active indicators are tied to a source and 111 carry data through the whole pipeline; the busiest live series is an hourly air-quality feed at 20,069 points, and history pulled during backfill reaches as far back as the year 2000. Outcomes by source type appear in Table II, and the status machine with production counts in Figure 6.

B. Self-healing effectiveness

43 wrappers, 10.9% of the total and split 26 spreadsheet, 14 API, and 3 CSV, threw an uncaught exception while running and were taken up by the repair loop, which shipped 50 redeployments; counted per wrapper, the repair depth was 37 at a single attempt, 5 at two, and 1 at three. "Recovery" can be read three ways, and they do not agree, so we keep them apart. Measured by recurrence, the same crash came back after the last repair for exactly one of the 43, so 42 stayed crash-free. Measured by final status, 39 of 43 (90.7%) closed in a non-error state. Measured by output, 32 resumed sending data after their last repair, 20,351 points, about an eighth of everything collected. The one wrapper that burned all three attempts never stabilised; it shows a non-error status only because its resource was deleted afterwards, and we do not score the deepest-retry case as a win. The repairs themselves were fast and untouched

by a person: a median 33.4 s from crash to redeployment (quartiles 24.2 and 50.8 s, 90th percentile 60.5 s, longest 80.4 s over the 50 events), up to 30 s of which is the detection cycle. The typical crash arrived 16.8 s into a run, in the fetch-and-parse stage, and none occurred while a poller waited idle.

The run also offers a pre-automation reference period. For the first 65 days no repair loop was active (it first appears in the logs on 2026-03-19), and over that stretch 36 crashes of this same exit class were simply left to die; we read the comparison descriptively, since the early period differs in source mix and load and is not a matched control. Two features of the timeline qualify the healing counts. Thirty-one of the 50 repair events land on a single day, when a credential fix set off a redeploy that restarted every wrapper together, so the events are correlated and the pooled latencies above reflect that batch more than 50 separate failures. And since each relaunch overwrites the stderr file, a triggering traceback survives only where the last run also crashed; for the wrappers that recovered we infer the trigger from the fixes their code carries, which means these counts describe the *repairs* the model made rather than independently observed causes. So read, a comment or guard pointing at a mended defect shows up in 7 of the 43 crashed wrappers and in none of the 353 that never crashed, and a timezone fix in 4 of 43 against 1 of 353. Table III sets the repair-depth breakdown beside the fault analysis.

TABLE III
WHY THE 69 `error`-STATUS WRAPPERS ENDED THERE, AND HOW THE REPAIR LOOP FARED OVER THE 43 CRASHED WRAPPERS AND THEIR 50 AUTOMATIC REPAIRS.

(a) Root causes of `error` status ($n = 69$)

Stage	Root cause	n
Generation	Model daily free-tier quota (HTTP 429)	9
Generation	Model per-request rate limit (HTTP 429)	1
Generation	Model provider overloaded (HTTP 503)	1
Generation	Service restart during generation	3
Runtime	Execution watchdog (3600 s), API pollers[a]	54
Runtime	Crash; repair blocked by LLM quota (429)	1
	Total	69

(b) Self-healing outcomes (43 wrappers, all crashes exit 1)

Healed after 1 repair (crash never recurred)	37
Healed after 2 repairs	5
Still crashing after 3 repairs (retries exhausted)	1
Final status: completed / stopped / executing	28 / 8 / 3
Final status: error (later watchdog timeout)[b]	4
Wrapper-level healing success	42/43
Median crash-to-redeploy repair latency	33.4 s

[a]stderr held no traceback (53 of 54 blank, one harmless warning); the kills happened during idle waits between polls, the one-hour watchdog being tighter than these feeds' daily or monthly periods. By then 48 of the 54 had shipped 52,372 points. [b]Each of the four was repaired and never crashed again; the later `error` comes from the watchdog limit, not the repaired code.

C. Repair case studies

Four wrappers span the range. The causes in the first two are read off the fix, the third has its exception quoted in the repaired file, and the fourth kept its trace because it never recovered.

1) Aware-versus-naive datetime comparison (API, healed): A yearly urban-waste API wrapper fell over 82 s into its backfill while matching timezone-aware dates from the source against timezone-naive window bounds. Its repair brings both bounds to naive form before the comparison and downgrades an empty fetch from an exception to a log line. Back in service after 34.7 s, it ran to completion with 596 points.

2) Doubled date parameters (API, healed): An air-quality wrapper had been given an endpoint URL that already pinned a date window; the generated code added a second one, and the server replied HTTP 400. The repair removes every variant of the start and end parameters from the supplied URL before applying its own window and swallows any leftover 400 rather than failing. Back after 51.3 s, it was still streaming at the snapshot, 395 points in.

3) Header text in the data rows (spreadsheet, healed): A cultural-heritage workbook repeated its localized year header (`"Ano"`) down among the values, so the wrapper died on `int("Ano")`; the repaired file quotes the exception, so the cause is recorded in the fix. The repair swaps raw integer casts for coercing conversions and discards non-numeric rows, and the same change turns up in three neighbouring wrappers mended the same night. Back after 61.6 s, it finished.

4) A missing second column (CSV, not healed): A search-interest export came through as one column, leaving no $(\text{time}, \text{value})$ pair to find. Three repairs (27.0, 33.0, and 51.8 s) tried delimiter sniffing and bad-line skipping, but none could conjure a second column, and the loop stopped. A sibling wrapper reading a two-column variant of the very same export healed on its first try. The manager mends code; it cannot supply data the source withholds.

D. Faults inside and outside the manager

Table III sorts the 69 wrappers that ended in `error` by cause, and few of them indict the generated code. Fourteen never reached the running stage: nine ran into the model's daily free-tier quota, one a per-request rate limit, one a provider overload, and three more were cut off mid-generation when a service restarted. Another, larger set of 54 are API pollers the 3600 s watchdog stopped between polling cycles, 53 of them leaving no traceback; here the watchdog is just shorter than the source's daily or monthly rhythm, a setting mismatched to the feed rather than broken code, and 48 of these had already delivered 52,372 points before being cut short. One wrapper crashed and could not be mended because the repair call itself struck the quota ceiling, and that is the lone error-status case where real generated code went unrepaired. The split is the scope of control of Section IV-E made concrete: code defects and formatting quirks come off the operator's plate, while spent quota, bad credentials, and limits set for the wrong cadence stay outside the managed element and stay manual.

E. Pipeline integrity and throughput

Point counts thin out down the pipeline in a way that reconciles: the 164,322 wrappers emit become 148,131 raw and then 106,442 consolidated points in the resource store, and 100,550 points in the indicator store that collapse, after chronological merging, into series of 84,431. The losses trace to deleted resources (198 of 396, 126 of which left their wrapper `stopped`), to regeneration wiping earlier data, to points landing for resources with no indicator attached, and to timestamp-level deduplication at merge time. Where we could check integrity, it held: across 30 days the collector rejected nothing on schema grounds and raised only 18 duplicate-timestamp warnings, each settled by averaging; over an 8.3-hour control-plane window the resource service emitted 153 messages and the indicator service took in exactly 153, none lost in the broker; and the gateway handled 55,086 requests in about 20 days (near 2,750 a day, 80.3% of them HTTP 200). Resumption proved itself in the field: a container rebuild a few hours before the snapshot sent all 16 running wrappers through a restart, 15 of them continuing from stored watermarks while the one missing a high watermark re-read its history from scratch.

VI. Discussion

Seen as an autonomic manager, the deployment explains both its successes and its ceiling. Source-specific code stays out of human hands because generation is fenced in by a fixed contract, because the plan is grounded in what the source actually returns instead of what it is assumed to return, and because repair fires on a genuine production trace. The ceiling is precisely the faults that sit outside the managed element: the single error tied to unrepaired code was waiting on a quota, and the single crash the loop could not mend was over data that was not there.

The evidence has limits worth stating plainly. It is observational, drawn from one deployment, one model family, and one region, with no side-by-side control; the pre-automation period is a reference rather than a matched arm, and the crash events are correlated, not independent. We have no hand-written wrappers for the same sources to measure against, because none were ever built. We score a wrapper as recovered once it runs cleanly again, which says nothing about whether the values it extracts are right, since the collector inspects each batch's shape and never the underlying truth; and for the healed wrappers the cause is read from the repair because the relaunch destroys the trace, so the fault analysis characterizes the model's fixes more than independently observed failures. The floating `gemini-flash-latest` alias pins the model, so its underlying weights may have shifted across the run; the generation latency includes queueing and so overstates model time; the memory figure is an idle reading; and watermark coverage spans are computable for only the 11 wrappers whose low mark was kept. Two systemic risks deserve their own mention and remain open: the model quota is a single point of failure already behind 10 of the 69 terminal errors, and running model-written code as supervised production processes is contained here only by the semaphore and watchdog, not by any stronger sandbox. Whether the pattern carries beyond 396 wrappers and one region is argued from its design, not demonstrated.

VII. Conclusion

A smart city observatory turns source-agnostic once its wrappers look after themselves. Treating each wrapper as the element of one MAPE-K manager pulls three usually-separate behaviours into a single account: the manager plans a wrapper from metadata and a sample, runs it as a supervised process that resumes from its check-points, and re-plans it from production traces when it breaks, with a shared knowledge base holding the fixed contract and the repair rules it has accumulated. Across 149 days it kept 396 wrappers and 164,322 points flowing; of the 43 that crashed, all but one were returned to a crash-free state by no more than three repairs, each a median 33.4 s and with nobody in the loop. What it could not absorb was environmental, which sets the agenda: checking extracted values for meaning, and folding quota and credential faults into the manager as state it owns rather than leaving them on an operator's desk.

References

[1] G. Wiederhold, "Mediators in the architecture of future information systems," *Computer*, vol. 25, no. 3, pp. 38–49, 1992.

[2] N. Kushmerick, "Wrapper induction: Efficiency and expressiveness," *Artificial Intelligence*, vol. 118, no. 1–2, pp. 15–68, 2000.

[3] K. Lerman, S. N. Minton, and C. A. Knoblock, "Wrapper maintenance: A machine learning approach," *Journal of Artificial Intelligence Research*, vol. 18, pp. 149–181, 2003.

[4] J. O. Kephart and D. M. Chess, "The vision of autonomic computing," *Computer*, vol. 36, no. 1, pp. 41–50, 2003.

[5] IBM Corporation, "An architectural blueprint for autonomic computing," IBM, White Paper, 2005.

[6] M. C. Huebscher and J. A. McCann, "A survey of autonomic computing—degrees, models, and applications," *ACM Computing Surveys*, vol. 40, no. 3, pp. 1–28, 2008.

[7] U. Gretzel, M. Sigala, Z. Xiang, and C. Koo, "Smart tourism: foundations and developments," *Electronic Markets*, vol. 25, no. 3, pp. 179–188, 2015.

[8] D. Buhalis and A. Amaranggana, "Smart tourism destinations," in *Information and Communication Technologies in Tourism 2014*, Z. Xiang and I. Tussyadiah, Eds. Cham: Springer International Publishing, 2013, pp. 553–564.

[9] European Commission, Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs, "The European Tourism Indicator System: ETIS toolkit for sustainable destination management," Publications Office of the European Union, Luxembourg, Tech. Rep., 2016.

[10] P. Modica, A. Capocchi, I. Foroni, and M. Zenga, "An assessment of the implementation of the European Tourism Indicator System for sustainable destinations in Italy," *Sustainability*, vol. 10, no. 9, p. 3160, 2018.

[11] M. Fuchs, W. Höpken, and M. Lexhagen, "Big data analytics for knowledge generation in tourism destinations – A case from Sweden," *Journal of Destination Marketing & Management*, vol. 3, no. 4, pp. 198–209, 2014.

[12] J. A. Ivars-Baidal, M. A. Celdrán-Bernabeu, F. Femenia-Serra, J. F. Perles-Ribes, and D. Giner-Sánchez, "Measuring the progress of smart destinations: The use of indicators as a management tool," *Journal of Destination Marketing & Management*, vol. 19, p. 100531, 2021.

[13] J. A. Ivars-Baidal, J. F. Vera-Rebollo, J. Perles-Ribes, F. Femenia-Serra, and M. A. Celdrán-Bernabeu, "Sustainable tourism indicators: what's new within the smart city/destination approach?" *Journal of Sustainable Tourism*, vol. 31, no. 7, pp. 1556–1582, 2023.

[14] M. Chen, J. Tworek, H. Jun, Q. Yuan, H. P. de Oliveira Pinto, J. Kaplan, H. Edwards, Y. Burda, N. Joseph, G. Brockman, A. Ray, R. Puri, G. Krueger, M. Petrov, H. Khlaaf, G. Sastry, P. Mishkin, B. Chan, S. Gray, N. Ryder, M. Pavlov, A. Power, L. Kaiser, M. Bavarian, C. Winter, P. Tillet, F. P. Such, D. Cummings, M. Plappert, F. Chantzis, E. Barnes, A. Herbert-Voss, W. H. Guss, A. Nichol, A. Paino, N. Tezak, J. Tang, I. Babuschkin, S. Balaji, S. Jain, W. Saunders, C. Hesse, A. N. Carr, J. Leike, J. Achiam, V. Misra, E. Morikawa, A. Radford, M. Knight, M. Brundage, M. Murati, K. Mayer, P. Welinder, B. McGrew, D. Amodei, S. McCandlish, I. Sutskever, and W. Zaremba, "Evaluating large language models trained on code," *arXiv preprint arXiv:2107.03374*, 2021. [Online]. Available: https://arxiv.org/abs/2107.03374

[15] C. S. Xia, Y. Wei, and L. Zhang, "Automated program repair in the era of large pre-trained language models," in *45th IEEE/ACM International Conference on Software Engineering, ICSE 2023, Melbourne, Australia, May 14-20, 2023*. IEEE, 2023, pp. 1482–1494. [Online]. Available: https://doi.org/10.1109/ICSE48619.2023.00129

[16] A. Madaan, N. Tandon, P. Gupta, S. Hallinan, L. Gao, S. Wiegreffe, U. Alon, N. Dziri, S. Prabhumoye, Y. Yang, S. Gupta, B. P. Majumder, K. Hermann, S. Welleck, A. Yazdanbakhsh, and P. Clark, "Self-Refine: Iterative refinement with self-feedback," in *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10-16, 2023*, 2023. [Online]. Available: http://papers.nips.cc/paper_files/paper/2023/hash/91edff07232fb1b55a505a9e9f6c0ff3-Abstract-Conference.html

[17] N. Shinn, F. Cassano, A. Gopinath, K. Narasimhan, and S. Yao, "Reflexion: Language agents with verbal reinforcement learning," in *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10-16, 2023*, 2023. [Online]. Available: http://papers.nips.cc/paper_files/paper/2023/hash/1b44b878bb782e6954cd888628510e90-Abstract-Conference.html

[18] X. Chen, M. Lin, N. Schärli, and D. Zhou, "Teaching large language models to self-debug," in *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024. [Online]. Available: https://openreview.net/forum?id=KuPixIqPiq

[19] T. X. Olausson, J. P. Inala, C. Wang, J. Gao, and A. Solar-Lezama, "Is self-repair a silver bullet for code generation?" in *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024. [Online]. Available: https://openreview.net/forum?id=y0GJXRungR

[20] S. Arora, B. Yang, S. Eyuboglu, A. Narayan, A. Hojel, I. Trummer, and C. Ré, "Language models enable simple systems for generating structured views of heterogeneous data lakes," *Proceedings of the VLDB Endowment*, vol. 17, no. 2, pp. 92–105, 2023.

[21] W. Huang, Z. Gu, C. Peng, J. Liang, Z. Li, Y. Xiao, L. Wen, and Z. Chen, "AutoScraper: A progressive understanding web agent for web scraper generation," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Y. Al-Onaizan, M. Bansal, and Y.-N. Chen, Eds. Miami, Florida, USA: Association for Computational Linguistics, Nov. 2024, pp. 2371–2389. [Online]. Available: https://aclanthology.org/2024.emnlp-main.141/

[22] A. Narayan, I. Chami, L. Orr, and C. Ré, "Can foundation models wrangle your data?" *Proceedings of the VLDB Endowment*, vol. 16, no. 4, pp. 738–746, 2022.

[23] H. Zhang, Y. Dong, C. Xiao, and M. Oyamada, "Jellyfish: Instruction-tuning local large language models for data preprocessing," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Miami, Florida, USA: Association for Computational Linguistics, Nov. 2024, pp. 8754–8782.

[24] V. Crescenzi, G. Mecca, and P. Merialdo, "RoadRunner: Towards automatic data extraction from large web sites," in *Proceedings of the 27th International Conference on Very Large Data Bases (VLDB)*. Morgan Kaufmann, 2001, pp. 109–118.

[25] A. H. F. Laender, B. A. Ribeiro-Neto, A. S. da Silva, and J. S. Teixeira, "A brief survey of web data extraction tools," *SIGMOD Record*, vol. 31, no. 2, pp. 84–93, 2002.

[26] E. Ferrara, P. De Meo, G. Fiumara, and R. Baumgartner, "Web data extraction, applications and techniques: A survey," *Knowledge-Based Systems*, vol. 70, pp. 301–323, 2014.

[27] N. Kushmerick, "Wrapper verification," *World Wide Web*, vol. 3, no. 2, pp. 79–94, 2000.

[28] P. Vassiliadis, A. Simitsis, and S. Skiadopoulos, "Conceptual modeling for ETL processes," in *Proceedings of the 5th ACM International Workshop on Data Warehousing and OLAP (DOLAP '02)*. McLean, VA, USA: ACM, 2002, pp. 14–21.

[29] P. Vassiliadis, "A survey of Extract–Transform–Load technology," *International Journal of Data Warehousing and Mining*, vol. 5, no. 3, pp. 1–27, 2009.

[30] E. Rahm and P. A. Bernstein, "A survey of approaches to automatic schema matching," *The VLDB Journal*, vol. 10, no. 4, pp. 334–350, 2001.

[31] M. Stonebraker, D. Bruckner, I. F. Ilyas, G. Beskales, M. Cherniack, S. B. Zdonik, A. Pagan, and S. Xu, "Data curation at scale: The Data Tamer system," in *Sixth Biennial Conference on Innovative Data Systems Research, CIDR 2013, Asilomar, CA, USA, January 6-9, 2013, Online Proceedings*. www.cidrdb.org, 2013. [Online]. Available: http://cidrdb.org/cidr2013/Papers/CIDR13_Paper28.pdf

[32] T. Furche, G. Gottlob, L. Libkin, G. Orsi, and N. W. Paton, "Data wrangling for big data: Challenges and opportunities," in *Proceedings of the 19th International Conference on Extending Database Technology (EDBT 2016)*. Bordeaux, France: OpenProceedings.org, 2016, pp. 473–478.